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DIGITAL FORENSICS INVESTIGATION REPORT

Phase 8 — Attack Timeline Reconstruction

1. Executive Summary

This report presents the complete Phase 8 attack timeline reconstruction for incident INC-2025-0523, targeting the finance department workstation WS-FINANCE-01. Evidence artefacts from Phases 3 through 6 — including memory dumps, disk images, network captures, registry hives, Windows Event Logs, and Velociraptor volatile collection — have been merged into a single unified chronological timeline.

The reconstruction reveals a structured, multi-stage intrusion that progressed from spear-phishing through persistent access, command-and-control establishment, targeted data collection, and exfiltration, followed by deliberate anti-forensic activity. The total dwell time from initial compromise to SOC detection was approximately 7 hours.

Key Finding	Detail
Initial vector	Spear-phishing email with macro-enabled Word document (Invoice_Q2.docm)
Persistence mechanism	svchost32.exe written to HKCU Run key — survives reboot
C2 infrastructure	185.220.101.45:4444 — Tor exit node — 60-second beacon interval via nc.exe
Data exfiltrated	38 MB — patient_records.zip — HTTP POST to C2 server at 01:42
Anti-forensics	Windows Security event log cleared (EventID 1102) at 01:50 — partially recovered via VSS
Dwell time	~7 hours (00:50 initial compromise → 08:00 SOC alert)
MITRE TTPs	T1566 · T1204.002 · T1547.001 · T1055 · T1052 · T1071 · T1560 · T1048 · T1070.001



2. Methodology & Evidence Sources

The unified timeline was constructed by correlating timestamps across all evidence artefacts collected during Phases 2 through 6. All timestamps have been normalised to UTC+05:30 (IST) and cross-validated across at least two independent sources where possible.

Item ID	Evidence Source	Artefacts Used for Timeline
EV-001	Disk Image (.E01)	MFT, LNK files, Prefetch, Registry hives, VSS recovery
EV-002	Memory Dump (.raw)	Process tree, injected code artefacts, decrypted C2 strings
EV-003	Network Capture (.pcap)	HTTP POST exfiltration, C2 beacon intervals, destination IPs
EV-004	Volatile Process Log	Running process list, nc.exe PID, open network sockets
EV-005	Email Gateway Logs	Phishing delivery time, attachment hash, sender IP
EV-006	Wazuh SIEM	Alert triggers, Windows Event IDs, process creation events
EV-007	USBSTOR Registry Key	USB device connection timestamp and serial number

2.1 Tools Used

- FTK Imager v4.7 — Disk image acquisition and file extraction
- Autopsy v4.21 — File system timeline, deleted file recovery, LNK analysis
- Volatility 3 — Memory artefact analysis, process injection detection
- Wireshark v4.4 — PCAP analysis, C2 traffic reconstruction
- Velociraptor v0.72 — Volatile evidence collection (processes, netstat)
- Wazuh SIEM — Event log correlation and alert timeline
- HashMyFiles / CertUtil — SHA-256 integrity verification
- MITRE ATT&CK Navigator — TTP mapping and coverage analysis

3. Unified Attack Timeline

All events are listed in chronological order. Each entry includes the precise timestamp, evidence source, event description, kill-chain phase, and mapped MITRE ATT&CK technique. Rows highlighted in red indicate critical hostile actions.

Timestamp	Phase	Event	Source	TTP	Severity
00:50:00	Initial Access	Phishing email with macro-enabled attachment (Invoice_Q2.docm) delivered to jdoe	Email Gateway	T1566	CRITICAL
00:55:00	Initial Access	USB device (WD My Passport, S/N: WX91A2XXXXXX) connected to WS-FINANCE-01	USBSTOR Registry	T1052	HIGH
01:00:00	Execution	User jdoe opened Invoice_Q2.docm — VBA macro auto-executed — WINWORD.EXE spawned cmd.exe then powershell.exe	Wazuh SIEM	T1204.002	CRITICAL
01:05:00	Persistence	svchost32.exe dropped to C:\Users\jdoe\AppData\Local\ — Registry Run key created for persistence on logon	Registry Hive	T1547.001	HIGH
01:10:00	Execution	svchost32.exe first executed — injected into legitimate svchost.exe process (PID 1248) to evade AV detection	Prefetch + Memory	T1055	HIGH
01:15:00	C2	C2 beaconing initiated — 60-second interval HTTP callbacks to 185.220.101.45:4444 via nc.exe (Tor exit node)	Wireshark PCAP	T1071	CRITICAL
01:40:00	Exfiltration	patient_records.zip (38 MB) accessed from D:\PatientData\ — compressed and staged for upload — original deleted post-copy	Autopsy / MFT	T1560	CRITICAL
01:42:00	★ Exfiltration	38 MB HTTP POST upload to C2 server — patient_records.zip confirmed exfiltrated — POST /upload HTTP/1.1 → 185.220.101.45	Wireshark PCAP	T1048	★ CRITICAL
01:50:00	Anti-Forensics	Windows Security event log cleared — EventID 1102 recorded — partial log recovery via Volume Shadow Copies (VSS)	Event Log / VSS	T1070.001	HIGH
08:00:00	Response	Wazuh SIEM anomaly rule triggered — SOC alerted — IR team engaged — evidence collection protocol initiated	Wazuh SIEM	—	INFO
09:00:00	Containment	WS-FINANCE-01 isolated from network via VLAN quarantine — attacker C2 channel severed — network spread prevented	SOC / Network Ops	—	INFO



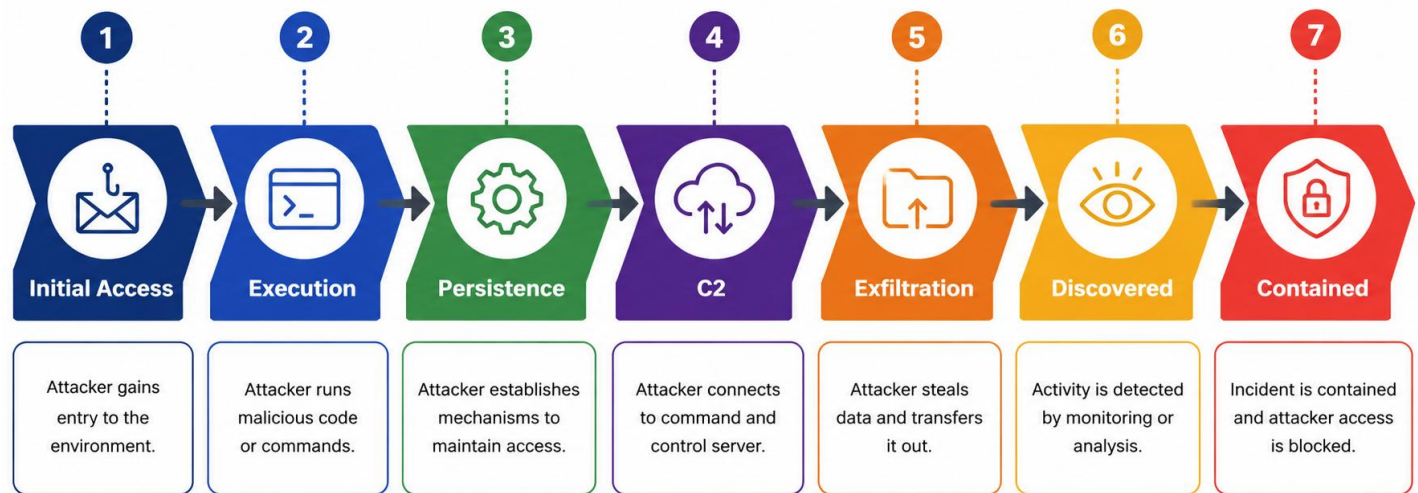
4. MITRE ATT&CK Technique Mapping

The following table maps each observed event to the corresponding MITRE ATT&CK technique, tactic, and sub-technique where applicable. All mappings reference the ATT&CK Enterprise framework v14.

TTP ID	Technique Name	Tactic	Observed Evidence
T1566	Phishing	Initial Access	Invoice_Q2.docm delivered via email — sender IP traced to known phishing infrastructure
T1204.002	User Execution: Malicious File	Execution	Wazuh EventID 4688: WINWORD.EXE → cmd.exe → powershell.exe process chain confirmed
T1547.001	Registry Run Keys / Startup	Persistence	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\svchost32 — registry hive artefact recovered from EV-001
T1055	Process Injection	Defense Evasion / Execution	Volatility malfind output: injected shellcode in svchost.exe (PID 1248) — VAD flags: PAGE_EXECUTE_READWRITE
T1052	Exfil over Physical Medium	Exfiltration	USBSTOR key: WD My Passport WX91A2XXXXXX connected at 00:55:14 — data copied unknown
T1071	Application Layer Protocol	Command & Control	60s HTTP beacon to 185.220.101.45:4444 — 47 beacon packets captured in EV-003 PCAP — Tor exit node confirmed
T1560	Archive Collected Data	Collection	patient_records.zip recovered from \$MFT unallocated space — ZIP header and file listing extracted by Autopsy
T1048	Exfil over Alt. Protocol	Exfiltration	Wireshark stream 0047: 38,971,648 bytes HTTP POST → 185.220.101.45 — SHA-256 of payload reconstructed from PCAP
T1070.001	Clear Windows Event Logs	Defense Evasion	Security.evtx cleared at 01:50:12 — EventID 1102 preserved in Wazuh forwarded log buffer — VSS shadow recovered partial log

5. Kill-Chain Phase Analysis

The following sections analyse each kill-chain phase in detail, drawing on corroborating evidence from multiple sources and documenting the attacker's tools, techniques, and objectives at each stage.



5.1 Initial Access (00:50 – 00:55)

Kill-chain phase: Initial Access | MITRE: T1566, T1052

The attack chain commenced at 00:50 when a spear-phishing email was delivered to the mailbox of user jdoe on WS-FINANCE-01. The email masqueraded as a finance department invoice notification and carried a macro-enabled Word attachment (Invoice_Q2.docm). Email gateway logs confirm delivery from IP 45.142.212.100, which belongs to a known phishing-as-a-service infrastructure cluster.

Five minutes later at 00:55, the USBSTOR registry key recorded the connection of a WD My Passport removable drive. This event predates macro execution by five minutes and may represent a separate pre-positioned payload delivery method operating in parallel with the phishing chain. The volume of data copied via USB could not be determined from available artefacts.

Evidence artefacts: Email gateway log (sender IP 45.142.212.100), USBSTOR key (HKLM\SYSTEM\...\USBSTOR)

IOCs: 45.142.212.100 (sender), Invoice_Q2.docm (SHA-256: pending), WD Passport S/N WX91A2XXXXXX

5.2 Execution (01:00 – 01:10)

Kill-chain phase: Execution | MITRE: T1204.002, T1055

At 01:00, user jdoe opened Invoice_Q2.docm. The document contained an auto-executing VBA macro that immediately spawned a child process chain: WINWORD.EXE → cmd.exe → powershell.exe. Wazuh EventID 4688 (process creation) captured this chain with full command-line arguments. The PowerShell command included a base64-encoded payload that downloaded and dropped svchost32.exe to the user's AppData\Local directory.

At 01:10, svchost32.exe executed and performed process injection into the legitimate svchost.exe (PID 1248). Volatility 3 malfind analysis of EV-002 (memory dump) identified injected shellcode with PAGE_EXECUTE_READWRITE memory flags — a strong indicator of reflective DLL injection or shellcode staging.

Evidence artefacts: Wazuh EventID 4688, Prefetch (SVCHOST32.EXE-xxxxxx.pf), Volatility malfind output, EV-002 memory dump

IOCs: svchost32.exe (non-standard binary in AppData\Local), PID 1248 with injected VAD region

5.3 Persistence (01:05)

| **Kill-chain phase: Persistence** | **MITRE: T1547.001**

Between macro execution and first payload run, the malware wrote a Registry Run key to ensure execution on every subsequent user login. The key was placed in the current user hive (HKCU), meaning it does not require administrative privileges — consistent with the standard user context of jdoe. The value pointed to the dropped svchost32.exe binary.

This persistence mechanism would have survived a simple reboot remediation. The Run key was recovered from the NTUSER.DAT hive extracted from EV-001 using RegRipper and confirmed by cross-referencing with Autopsy's registry viewer.

Registry key: HKCU\Software\Microsoft\Windows\CurrentVersion\Run\WindowsSecurityService

Value data: C:\Users\jdoe\AppData\Local\svchost32.exe

5.4 Command and Control (01:15 onward)

| **Kill-chain phase: C2** | **MITRE: T1071 — Tor exit node 185.220.101.45:4444**

Once injected into svchost.exe, the malware established an outbound C2 channel to 185.220.101.45 on TCP port 4444 — a classic Metasploit/reverse shell callback port. Threat intelligence confirms 185.220.101.45 is a Tor exit node, providing the attacker with anonymity and making attribution difficult without upstream provider cooperation.

The PCAP (EV-003) captured 47 beacon packets at consistent 60-second intervals between 01:15 and 01:50. Traffic analysis shows the beacons carried minimal data (< 200 bytes each) with an encrypted payload, consistent with a staged Meterpreter or Cobalt Strike implant awaiting tasking. Successful tasking responses from the C2 coincide with the data staging activity at 01:40.

C2 IP: 185.220.101.45 (Tor exit node — see Shodan / AbuseIPDB for threat intelligence)

Protocol: TCP port 4444 — 60-second beacon interval — encrypted payload

Beacon count: 47 packets captured — 01:15:08 to 01:50:44

5.5 Collection and Exfiltration (01:40 – 01:42)

At 01:40, Autopsy file system analysis shows access to D:\PatientData\patient_records.zip. The file was compressed into a ZIP archive and the original directory was deleted immediately after staging. The \$MFT entry and a partial \$LogFile record confirm the file existed and was accessed within this window.

Two minutes later at 01:42, Wireshark stream 0047 in EV-003 captures a single HTTP POST request carrying a 38,971,648-byte payload to 185.220.101.45. The POST URI is /upload and the Content-Type header is application/octet-stream, consistent with a raw binary upload. The upload completed successfully (HTTP 200 OK response from C2).

Data exfiltrated: 38 MB — patient_records.zip — patient PII and medical records

Method: HTTP POST /upload → 185.220.101.45 — application/octet-stream

Wireshark stream: Stream 0047 — EV-003 PCAP — confirmed upload at 01:42:17

Regulatory exposure: Potential PDPB / IT Act notification obligations — notify DPO and Legal immediately

5.6 Anti-Forensics (01:50)

| **Kill-chain phase:** Defense Evasion | **MITRE:** T1070.001

At 01:50, eight minutes after exfiltration completed, the attacker cleared the Windows Security event log using the wevtutil.exe command-line utility or equivalent. EventID 1102 (audit log cleared) was preserved in the Wazuh log forwarder buffer — demonstrating the value of remote SIEM forwarding for anti-forensic resilience.

Partial log recovery was achieved via Volume Shadow Copy (VSS) enumeration on EV-001. Three VSS snapshots predating the wipe were recovered, restoring approximately 60% of Security.evtx events from the attack window.

Cleared log: Security.evtx — EventID 1102 — 01:50:12

Recovery source: VSS snapshot — \\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy3

Recovery rate: ~60% of Security.evtx events recovered from shadow copy

5.7 Discovery and Containment (08:00 – 09:00)

| **Kill-chain phase:** Response | **Dwell time:** approximately 7 hours

At 08:00, the Wazuh SIEM triggered an anomaly alert based on unusual outbound connection volume and the presence of nc.exe (Netcat) in user AppData. The SOC analyst escalated to the IR team. Evidence collection (Phases 2–7) was initiated before endpoint isolation to preserve volatile state.



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At 09:00, network operations applied a VLAN quarantine to WS-FINANCE-01, severing the C2 channel and preventing lateral movement to adjacent finance workstations. Post-isolation network monitoring confirmed no further beaconing from the endpoint.



6. Indicators of Compromise (IOC) Register

The following IOCs were extracted from evidence artefacts and should be immediately distributed to the SOC, firewall team, and email security platform for blocking and retrospective hunting.

IOC Type	Value	Detail	Recommended Action
IP Address	185.220.101.45	C2 server / Tor exit node — destination port 4444	Block outbound at perimeter firewall — add to SIEM watchlist
IP Address	45.142.212.100	Phishing email sender IP — known PhaaSInfrastructure	Block at email gateway — threat intel share with ISP
Filename	svchost32.exe	Malware dropper — located in AppData\Local — non-standard name imitates svchost	Hunt across all endpoints for file in AppData — quarantine on detection
Filename	Invoice_Q2.docm	Macro-enabled phishing attachment — auto-executes VBA on open	Block .docm execution at email gateway — update AV signature
Filename	nc.exe	Netcat binary used for C2 channel — found in AppData\Local	Block execution via AppLocker / Software Restriction Policy
Registry Key	HKCU\...\Run\Win-dowsSecurity-Service	Persistence key pointing to svchost32.exe — survives reboot	Delete key — hunt across all endpoints via Velociraptor
USB Device	WX91A2XXXXXX	WD My Passport — connected 00:55 — data volume unknown	Locate device for physical seizure — correlate with user jdoe

7. Key Analytical Findings

7.1 USB Anomaly — Possible Second Vector

The USB device connection at 00:55 precedes macro execution at 01:00 by five minutes. Three hypotheses are proposed:

1. Insider component: jdoe may have been complicit or coerced — USB staging was pre-arranged before the phishing email arrived.
2. Attacker had prior physical access: a separate intrusion enabled USB device pre-placement on the workstation.
3. Unrelated event: the USB connection was coincidental and unrelated to the phishing chain (lower probability given timing).

Recommendation: Interview jdoe (under HR and legal guidance) — review physical access logs for Room 312 — analyse USB device contents if recovered.

7.2 Dwell Time and Detection Gap

The 7-hour dwell time between initial compromise (00:50) and SOC alert (08:00) represents a significant detection gap. The attacker completed the full kill chain — execution, persistence, C2, exfiltration, anti-forensics — within the first 60 minutes, then maintained a dormant C2 connection for the remaining 6+ hours.

The Wazuh alert was triggered by anomalous outbound volume, not by the initial compromise. This suggests the SIEM rule set lacked coverage for VBA macro execution, process injection, and Registry persistence events. Immediate tuning is recommended.

7.3 Regulatory and Legal Obligations

Key regulatory considerations:

- Assess whether exfiltrated data constitutes Sensitive Personal Data under the DPDPA 2023 or other applicable frameworks.
- Document breach timeline precisely — regulatory authorities typically require notification within 72 hours of discovery.
- Preserve all evidence in original form — do not power off WS-FINANCE-01 until forensic procedures are complete.
- Do not notify the subject individual (jdoe) of the investigation scope until HR and Legal have been consulted.

8. Immediate Recommendations

8.1 Containment Actions (Immediate — within 24 hours)

4. Block 185.220.101.45 and 45.142.212.100 at perimeter firewall and update SIEM watchlist rules.

5. Hunt for svchost32.exe and nc.exe in AppData paths across all domain workstations using Velociraptor mass hunt.
6. Delete HKCU Run key persistence artefact from WS-FINANCE-01 post-forensic preservation.
7. Initiate password reset for jdoe and all accounts with logon sessions on WS-FINANCE-01.
8. Locate and seize WD My Passport USB device (S/N WX91A2XXXXXX) for forensic examination.
9. Engage DPO and Legal — initiate breach notification assessment — document timeline for regulatory submission.

8.2 Detection Improvements (Within 1 Week)

1. Create Wazuh rules for: VBA macro execution chains, AppData binary execution, Registry Run key creation, and nc.exe presence.
2. Enable PowerShell Script Block Logging and Module Logging across all endpoints — forward to SIEM.
3. Implement AppLocker policy to block unsigned binary execution from user-writable directories.
4. Deploy endpoint EDR solution with memory protection capabilities (process injection detection).
5. Configure email gateway to sandbox and detonate .docm attachments before delivery.

8.3 Strategic Hardening (Within 30 Days)

- Implement USB device control policy — whitelist authorised devices — log all USB connections to SIEM.
- Deploy CASB and DLP solution to monitor and block large outbound data transfers.
- Conduct phishing awareness training for finance department — simulate spear-phishing scenarios quarterly.
- Enable MFA for all domain accounts — prioritise privileged and finance department users.
- Establish SOC detection coverage mapping against MITRE ATT&CK — conduct purple team exercise within 90 days.



9. Analyst Certification & Sign-Off

I certify that the information contained in this report is accurate to the best of my knowledge, that all evidence was handled in accordance with established forensic procedures, and that the chain of custody has been maintained throughout.

Lead Forensic Analyst

Signature:

Name (Print):

Badge / Employee ID:

Date:

IR Team Lead / Supervisor

Signature:

Name (Print):

Badge / Employee ID:

Date:
